



Dalhousie University

Sp

Department of Psychology

**27th Annual
In-House Convention**

26-27 April 2001

In Memory



*Dr. Werner (Vern) K. Honig
June 20, 1932-April 17, 2001*

Welcome to the 2001 Psychology and Neuroscience In-House Convention

Welcome to the 27th Annual Dalhousie University Department of Psychology In-House Convention. This year's conference is being presented in memory of Dr. Werner (Vern) K. Honig.

Thanks should be extended to the In-House Convention Committee for the planning and organization. The committee consists of the following individuals: D. Bernier, J. Connolly, K. Devoulyte, K. Peace, L. Stanford, T. Walsh, and A. Wintink..

Friendly Reminders

- Please bring your own mug for coffee or tea!
- All presentations are listed in the name of the presenter from the abstracts submitted.
- There will be a social at the end of Thursday and pizza and beer at the end of Friday in the 2nd Floor Lounge.

Thursday, April 26, 2000 (LSC 4258)

9:00 **Introduction to the In-House Convention** —Richard Brown

9:30—10:30 **Session 1.**

- T1) Ann-Marie Penny ✕
- T2) Mary Ellen Large
- T3) Jingtian Wang
- T4) Elliott Marchant

10:30—10:45 **Break**

10:45—12:00 **Session 2.**

- T5) Steve Shaw
- T6) Pamela Loba ✕
- T7) Kyle Owens ✕
- T8) Dennis Phillips
- T9) Randy Newman

12:00— 1:45 **Lunch Break and Posters**

- P1) Kerri Purdy ✕
- P2) Lianne Stanford ✕
- P3) Boris Kablar ✕
- P4) Yannick Marchand ✕

1:45— 3:15 **Session 3.**

- T10) Amanda Wintink ✕
- T11) Heather Schellinck
- T12) Celeste Lefebvre ✕
- T13) Donald Mitchell
- T14) Jason Ivanoff
- T15) Trudi Walsh ✕ *Heather Schellinck*

3:15— 3:30 **Break**

3:30— 5:15 **Session 4.**

- T16) Will Brown
- T17) Susan Boehnke
- T18) Frank Elgar ✕
- T19) Jeff Hanock
- T20) Denise Bernier ✕
- T21) Cathy Forestell
- ~~T22~~) Ron Hoffman and Heather Schellinck

5:15— **Social in the 2nd Floor Psychology Lounge**

Friday, April 27, 2000 (LSC 4258)

9:00—10:30 **Session 5.**

T23) John Fisk -/

T24) John Fisk

T25) Sultan Darvesh

T26) Sarah Boudreau

T27) Will Brown

T28) Ryan D'Arcy

10:30—10:45 **Break**

10:45—12:00 **Session 6.**

T29) Marleen deGroot

T30) Kristine Devoulyte

T31) Nancy Garon

T32) Guy Gingras

T33) Doug Rasmussen

T34) John Christie

T35) J. Blustein

12:00—2:00 **Lunch Break and Posters**

P5) Lyana Sisca and Laura Reynolds

P6) Paddy McMullen

P7) Alicia Davis

P8) Angie Birt

P9) Naomi Doucette

2:00— 3:30 **Session 7.**

T36) Kazue Semba

T37) Amy Carr

T38) Sherry Stewart

T39) Richard Brown

T40) Peggy Ruyak

T41) Jennifer Theakston

3:30— 3:50 **Break**

3:50— 5:10 **Symposium—An Afternoon of Murder, Memory and Mayhem: Hosted by the Forensic Psychology Students**

—Opening Remarks

T42) Mike Woodworth

T43) Kris Peace

T44) Sean McCabe

T45) Paul Angelopoulos

T46) Lesley Spencer

T47) Mary Ann Campbell

—Closing Remarks

5:10— **Pizza and beer in the 2nd floor Psychology Lounge**

An Historical Introduction to the In-House Conference

Richard Brown

Almost everyone who has been in the Psychology Department over the last 26 years has given a presentation at the in-house conference. This introductory presentation will use photographs from the last 26+ years to stimulate memories of previous conferences. Photos of present and former faculty, students and PDF's in their most embarrassing moments will be presented.

T1. Thinking About ADHD: Is a Psychoeducational Perspective Useful in Understanding Attention-Deficit / Hyperactivity Disorder?

Daniel A. Waschbusch, Ann Marie Penny

The purpose of the proposed study is to investigate whether or not recent theories of intelligence are useful in understanding children with attention deficit hyperactivity disorder (ADHD). Participants will be 80 children aged 7 to 11, including children with and without ADHD, and with and without learning disorders. In order to assign each child a diagnosis, parents will complete a series of questionnaires assessing the child's behaviour, and each child will complete the Wide Range Achievement Test, third edition. Each child will also complete the Woodcock-Johnson Tests of Cognitive Abilities, third edition, which provides scores based on the Carroll-Horn-Cattell (CHC) theory of intelligence. These scores will be analyzed for patterns within diagnostic groups, as well as differences between groups.

T2. Priming effects of global and local attention on the categorization of objects: A pilot study.

Mary Ellen Large and Paddy McMullen

Images of objects are identified more quickly at the basic level (dog) than the subordinate level (beagle) because subordinate level identifications require more visual processing in order to differentiate exemplars who share many common features. The differences in speed of processing between basic and subordinate level categorizations may be explained by global (coarse aspects of structure) and local (fine detail) attentional differences. To investigate this possibility the priming effects of a global and local attentional task on the categorization of objects at the basic and subordinate level were examined. Additionally, distractors varied in their degree of similarity to the target. It was found that after priming at the local level participants responded more quickly to both basic and subordinate categorizations when the distractors were similar and after priming at the global level they responded more quickly to basic and subordinate categorizations when the distractors were dissimilar. This result suggested that an important distinction in the speed of processing for basic and subordinate categorizations is not the level of categorization per se but the visual context in which the categorizations occurs.

T3. The effects of the segmental and suprasegmental processes of spoken Chinese Mandarin on the auditory ERPs

Jingtian Wang and John F. Connolly

In this study, event-related brain potentials (ERPs) were used to investigate spoken Chinese Mandarin processes. Chinese four-character proverbs were presented auditorally to 15 native Mandarin speakers for listening comprehension. Five conditions used proverbs that ended with: 1) the accurate fourth character of the proverb; 2) a character with the same final vowel and tone as the correct proverb-terminal character, but different initial consonant and meaning; 3) a character with the same initial consonant and tone as the correct proverb-terminal character, but different final vowel and meaning; 4) a character with the same segments as the correct proverb-terminal character, but different tone and meaning; and, 5) a character with a totally different segments, tone and meaning from the correct ending character. The N400 was seen in the last four conditions with the largest N400 in the fifth condition while the smallest was seen in the fourth condition. Only a late positive component was found in the first condition. It was concluded that tone processing does play a role although not that much as segments in processing spoken Chinese. Before N400, there was another negativity whose peak varied across last four conditions, which suggested that within highly constrained proverb contexts, tone was processed earlier than vowel. In addition, the ERP scalp distribution varied not only across different conditions but also different time periods. These data suggest that an auditory code relies upon discrete processing stages with different neural mechanisms underlying consonant, vowel and tone processing. The findings are integrated into current theories of spoken word recognition.

T4. Sillier Cilia: The Hairy Walls of the Mouse Third Ventricle

Elliott Marchant, K. Devarajan, and B. Rusak

In the course of anatomical research aimed at confirming the results of microarray gene expression studies, we re-discovered the generally ignored fact that portions of the third ventricle in the hypothalamus are lined with cilia. We also made the novel observation that these cilia contain two proteins: the calcium-regulating protein parvalbumin, and the neurohormone oxytocin. Levels of both parvalbumin and oxytocin in the cilia were regulated both by time of day and by whether or not the mice had been exposed to light an hour earlier. The rapid responses of these proteins to light exposure suggest a dynamic regulation that may influence the secretion of oxytocin into the cerebrospinal fluid (CSF). Oxytocin has been implicated in various affiliative behaviours and in the regulation of excitatory neurotransmission in the brain. Our observations suggest novel mechanisms for rapid regulation of global brain function via hypothalamic modulation of hormonal secretion into the CSF.

RLH protein in the 3rd ventricle

for the first time - see also [ref. 1] [ref. 2]

see also [ref. 3] [ref. 4]

Marchant et al.

T5. Exchange of dye tracers across the blood-brain barrier systems around the fly's eye

Steve Shaw

To gauge the effectiveness of the barriers around the eye, quantitative estimates of dye exchange were made by microfluorometry, using sulforhodamine 101 (SR) or lucifer yellow (LY). Influx to the retina from the blood was calibrated by making a known injection into the retina, allowing influx and efflux to be measured on the same preparations. Efflux of SR averaged $2.8 \mu\text{M eye}^{-1} \text{ min}^{-1}$ per mM driving concentration of the dye injected, while the paired influx rate was significantly lower, $0.20 \mu\text{M eye}^{-1} \text{ min}^{-1}$ per mM dye injected, for an exchange ratio of 14:1. Values for LY were higher but not significantly different, for a ratio of 8:1. The efflux half-times for LY and SR also were not significantly different, 2.9-3.6 h, but were ~5 times slower than previous estimates. The molecular uptake rate for dye significantly exceeds that recently calculated for ATP usage during the maximum rate of neural signaling by the photoreceptors, 1000 bit s^{-1} (55×10^9 versus 7.5×10^9 molecules $\text{cell}^{-1} \text{ s}^{-1}$), and might represent the flux rate of the corresponding transporter. Earlier estimates of dye loss from the retina also used phenol red (PR). Photographic evidence indicates that with optimum concentrations in the eye, clearance of PR was also slow, taking 8-17 h. To identify the dye exchange route, $20 \mu\text{m}$ fluorescent microspheres were injected into the retina, based on reports that larger microspheres remain mobile in the eye and eventually to leave it. Instead, they penetrated a distal zone of mechanical weakness which opened up around the cone cells, then closed again, trapping the microspheres. The barrier sites which rate-limit influx and efflux to and from the eye and account for the flux asymmetry remain to be identified, and could be different neuroanatomical structures.

T6. Manipulations of the Features of Standard Video Lottery Terminal (VLT) Games: Lab-based and In-vivo Results

Pamela Loba, Sherry H. Stewart, Raymond M. Klein, & James R. Blackburn

A set of lab-based and in-vivo studies were conducted to identify game parameters that would reduce the risk of abuse of video lottery terminals (VLT's) by pathological gamblers, while exerting minimal effects on the behavior of non-pathological gamblers. Three manipulations of standard VLT game features were explored. Participants were exposed to: a counter which displayed a running total of money spent; a VLT spinning reels game where participants could no longer "stop" the reels by touching the screen; and sensory feature manipulations. In control conditions, participants were exposed to standard settings for either a spinning reels or a video poker game. For the lab-based study dependent variables were self-ratings of reactions to each set of parameters. A set of $2(3) \times 2 \times 2$ (game manipulation [experimental condition(s) vs. control condition] $\times$ game [spinning reels vs. video poker] $\times$ gambler status [pathological vs. non-pathological]) repeated measures ANOVA's were conducted on all dependent variables. The in-vivo study used behavioural ratings as the dependent measure. Findings for both studies suggest that the sensory manipulations (i.e., fast speed/sound or slow speed/no sound manipulations) produced the most robust reaction differences.

T7. The quickest cancellation in the west

Kyle C. Owens

Having the best tactic for successfully cancelling an action ensures the minimum of unwanted movements. Normal overt responses to environmental stimuli can unexpectedly require sudden cancellation. Strategies for movement cancellation include freezing all movement or making an alternate movement instead. It is not known which strategy is more effective at preventing an overt action. Participants can be instructed to cancel a response to a stimulus when presented with a 'stop-signal', or to replace the original response with an alternate action when presented with a 'change-signal'. Past research has investigated the effectiveness of inhibition mechanisms by plotting the likelihood of successful inhibition along points in time. This research has used stop-signals almost exclusively. Following a brief literature review, the present talk will outline a research proposal that is intended to provide a direct comparison of the effectiveness of inhibiting responses to stop-signals compared to change-signals. Preliminary data will be provided.

T8. Transient response of the central auditory nervous system directly recorded in unanesthetized animals

D.P. Phillips, S.E. Hall, Y.Q. Guo and R. Burkard

For 50 years, or more, people have studied the intensity-coding properties of the central auditory system by measuring neural response amplitude (or latency) as a function of the plateau level of a tone- or noise-burst stimulus. This is easy to do, because responses to such stimuli are often very transient and locked in time to the onset of the sounds; these features of the response provide opportunity to offer gainful employment to programmers or hardware companies who supply ways of summing or averaging those responses. In practice, we have been left with a – now long-standing – description of intensity coding in terms of sensitivity of the central auditory system's transient response to the plateau level of the stimulus. In this talk, we show that this conceptualization of intensity coding is in error, and we provide a more viable alternative.

T9. Identifying Phonological Awareness Deficits with Event-Related Brain Potentials (ERPs)

Randy Lynn Newman, John Connolly, Kimberly McIvor, Wayne MacDonald, and Sarah Arnold

Reading disabilities primarily arise from a deficit in phonological awareness, which is simply defined as the ability to identify, think about, or manipulate the individual sounds in words. The present research used event-related brain potentials (ERPs) to elucidate the neural correlates involved in phonological analysis, and how these differ in individuals with and without reading disabilities. Participants heard instructions, such as "Clap without the /c/ sound", where the target word (Clap) was always monosyllabic and began with a consonant cluster. The instructions were followed by presentation of a correct item (lap) or one of three incorrect items: (1) wrong consonant deleted (cap), (2) consonant cluster deleted (ap); or an (3) irrelevant word (goose). Those with efficient phonological skills elicited a Phonological Mismatch Negativity (PMN), a component associated with phonological expectations, to incorrect items and a P300, a component associated with memory updating, to correct items. Individuals with the greatest deficits in phonological skills showed no differentiation between correct and incorrect items, with the exception of a PMN to irrelevant words. Those with moderate deficits displayed a PMN to both irrelevant words and to items with the wrong consonant deleted. Further, these individuals elicited a P300 to both correct items, and to items with the consonant cluster deleted. Results will be discussed in terms of the capacity that ERP techniques have for identifying reading disabilities, and for assisting reading recovery programs.

T10. Kindling-induced emotionality in both male and female rats

Wintink, AJ, Young, N, Davis, AC, Gregus, A, Kalynchuk, LE.

Men and women with temporal lobe epilepsy often experience interictal (i.e. between seizure) emotional disturbances. Long-term kindling (i.e. repeated focal stimulations) in rats serves as a model to understand the nature of seizures and of interictal emotionality (i.e. anxious behavior in novel situations) but has not yet been used to investigate potential differences in emotionality between male and female rats. There is substantial reason to expect sex differences in interictal emotionality including clinical data showing that more women are diagnosed with anxiety related disorders than are men and the expression of anxiety can be manifested differently between men and women. In the present study, both short- (30 stimulations) and long- (99 stimulations) term kindling was used to assess differences in emotionality in both male and female rats. Rats were assessed on behavioral measures of anxiety and depression and sex differences were observed on some of the measures. The results provide the first examination of kindling-induced emotionality between male and female rats.

T11. Long-term effects of acute and chronic pain in infancy in mice

Heather M. Schellinck, Matthew Darrah and Lianne Stanford

Exposure to pain during development has long been hypothesized to disrupt behaviour in later life. (Hebb, 1949 Organization of Behaviour). We investigated the influence of early pain experience on anxiety and learning and memory in mice. Neonatal mice were exposed to an acute pain, i.e., a tail snip at postnatal day 8 or chronic pain, i.e., paw pricks on postnatal days 8-14. When tested in an elevated plus maze on day 30, both acute and chronic pain mice showed behavioural changes indicative of increased anxiety compared with control animals. The effects included a reduction in open arm entries, percent time spent in the open arms and a decrease in head dips. In addition, chronic pain animals showed a greater decrease in time spent in the open arms and in number of entries into the open arms as well as an increase in stretch attends compared with acute pain animals. There were no differences in the number of line crossings in the experimental and control animals. This pattern of results would suggest that the pain-exposed animals showed an overall increase in risk assessment, a behaviour associated with hypervigilance in humans. In contrast, when tested in the Morris Water Maze, these mice demonstrated an intact working memory in a three-day acquisition task as well as in a probe trial and reversal trials. Thus, it appears that their increased anxiety did not adversely effect their performance in a spatial working memory task. Overall, these results reflect the potential role of early experience in influencing behaviour in adult animals.

T12. An event-related brain potential (ERP) study of attention/working memory

Celeste Lefebvre, John Connolly and Gail Eskes

The purpose of this study is to adapt two commonly used standardized neuropsychological measures of attention/working memory for computer presentation and simultaneous ERP recordings. This study will investigate the effectiveness of ERPs to determine working memory capabilities, as measured by the Digit Span and Spatial Span tasks from the Wechsler Memory Scale-III, in a normal population of undergraduate students. Each subject will be administered the traditional and computerized version of one of the two tasks. We hypothesize that 1) an individual's results on the computerized version of the Digit Span or Spatial Span tasks will be comparable to their results on the task when it is traditionally administered 2) the ERP components will reflect attention/memory processes and can determine when a person has reached the capacity of their attention span/working memory and 3) the differential ERP response pattern will be observable on an individual subject level.

T13. Initial Recovery of Vision after Early Monocular Deprivation in Kittens is Faster When Both Eyes Are Open

Donald E. Mitchell, Guy Gingras and Peter C. Kind

The mild effects of binocular as compared to monocular visual deprivation led to the widely held view that the plasticity observed in the mammalian visual cortex during development reflects the operation of competitive interactions between the two eyes for synaptic space. More recent evidence suggests that inputs from the two eyes compete for limiting amounts of a retrograde molecule such as a neurotrophin, and active neurons can compete more efficiently than less active neurons. Competitive mechanisms, however, cannot explain the physiological and behavioural recovery in the deprived eye observed during a subsequent period of binocular vision in cats. Here we show, in direct comparisons of the speed of visual recovery in the deprived eye of kittens following a 6 day period of monocular deprivation, that recovery is initially faster in the situation where both eyes are open (binocular recovery) than with reverse lid-suture, where the animal is forced to employ the originally deprived eye. This state of affairs, which lasted for 4 - 8 days, is difficult to reconcile with accounts of synaptic plasticity based upon binocular competition but is consistent with homosynaptic mechanisms such as the model proposed by Binstock, Cooper and Munro (the BCM Model).

T14. Exogenous orienting without awareness depends on attentional control settings

Jason Ivanoff

Exogenous orienting, to a sudden-onset peripheral cue is thought to occur semi-automatically. Typically, exogenous orienting exhibits a biphasic pattern: responses to targets are initially facilitated by the cue but later they are inhibited (inhibition of return; IOR). Exogenous orienting to a cue is not fully automatic because it depends on the cue and target sharing a particular quality (e.g., sudden-onset). Thus, exogenous orienting is said to depend on the establishment of an attentional control setting. McCormick (1997) demonstrated that peripheral cues presented below a threshold of awareness attract attention and subsequently facilitate target processing. Interestingly, at late cue-target onset asynchronies he did not observe IOR, even though this is a robust effect often observed with exogenous cues. In the present study, masked and unmasked sudden-onset cues were presented under two conditions. In the first condition, the task was to respond rapidly to the onset of a target. Although the target task was the same in the second condition, the cue was detected and localized at the end of trial. IOR was present with and without masks in the first condition, but facilitation was not observed when the cues were masked. In the second condition, facilitation was observed for masked and unmasked cues, but IOR was not observed with masked cues. The results suggest that assessing cue awareness establishes a set for searching and this set gives rise to facilitation regardless of whether or not the cue can be consciously localized.

T15. Attachment Security, Maternal Sensitivity, and Children's Response to Pain: An Internal Working Model of Pain

Trudi Walsh

This study will examine the predictive validity of attachment security and pain behaviour following both everyday pain incidents and acute pain incidents. The present study will be conducted with a cohort of fifty, five-year-old children, followed since 6 months of age, and their mothers. The children's faces will be videotaped during their 5-year medically indicated inoculation shot. Children's pain reaction will be coded using the Children's Facial Coding Scale and parent-child interactions will be assessed with the Child Adult Medical Procedure Interaction Scale. Following the needle, both mothers and children will provide ratings of subjective pain intensity and pain behaviour. Mothers will also complete diaries following four everyday pain incidents that their child experiences. Included in the diary is information concerning the child's reaction and subsequent parental response. Quality of attachment will be assessed using a Separation-Reunion procedure, the Separation Anxiety Test, and the Pain and Relationship Task. It is hypothesized that attachment security classification will predict children's response to and behaviour following painful incidents. The mediating effect of maternal sensitivity, initially collected at both 6 and 18 months of age, will be also examined. Multimodal procedures will also be used to examine the role of third variables that may account for these potential relations. These will include receptive language, emotional regulation, and emotional awareness.

T16. Good genes, fluctuating asymmetry and human jealousy

Will Brown and Chris Moore

Ancestrally extrapair copulations (EPC) likely exerted adaptive pressures on hominids. One adaptive solution to reducing the probability of EPC is jealous emotions. Jealousy may be an adaptive strategy designed to retain one's mate. If jealousy was an ancestral mate retention tactic, then individuals more likely to be philandered upon should be more jealous. A predictor of likelihood to be philandered upon may be developmental instability. Developmental stability (DS) is a heritable indicator of an organism's ability to buffer environmental and genetic perturbances. DS is measured indirectly via fluctuating asymmetry (FA). FA is the deviation from perfect bilateral symmetry in normally bilateral traits. The less deviation from perfect symmetry is believed to be an intrinsically unfalsifiable indicator of 'good genes.' As predicted, more asymmetric individuals (as measured by digital calipers with .01mm accuracy) were more jealous. FA accounted for substantial proportion of variance in self-reported jealousy. Findings are consistent with jealousy being a condition-dependent mate retention tactic in humans.

T17. Listening for chaos amid order, or order amid chaos

Susan Boehnke, Susan Hall, Torsten Marquardt, Dennis Phillips

The listening brain generates auditory images with a location by examining the correlation between the sound waveforms that reach the two ears. If the sounds reaching the two ears are perfectly correlated (+ORDER) we perceive a compact sound image with a distinct location. If the two ears receive independent uncorrelated sounds (CHAOS), the perception is spatially diffuse and has no "location", like the noise of wind rustling leaves in a forest. We can digitally generate another, unnatural, type of perfectly correlated signal in which the signal phase at one ear is perfectly opposite to that of the signal reaching the other ear (-ORDER). This is perceptually somewhat compact, but doesn't have an easily specifiable "location". We have been examining the shortest detectable duration of each of these sounds when they occur between markers of another type: What is the shortest duration of ORDER you can detect in an ongoing stream of CHAOS, or of CHAOS in ORDER. It is much easier to detect the latter.

T18. Dieting and body dissatisfaction: An epidemiological survey of health behaviours in Canadian adolescents

Frank Elgar, Christine Clancy, Alan J. C. King

Identifying socio-demographic trends and psychological and behavioural correlates of dieting and body dissatisfaction (BD) in adolescents helps to identify those at greater risk of developing eating disorders. Analyses were performed on data collected from the Health Behaviours in School-Aged Children study in which 11,272 Canadian adolescents from grades 6 through 10 participated. Both BD and dieting were most prevalent among older and female adolescents, and BD increased the likelihood of dieting threefold. Other factors that increased the likelihood of dieting were depressed mood and low self-confidence. Adolescents who socialized more with peers, smoked more and consumed more alcohol were more likely to be dieting, suggesting peer influences on eating behaviours are pervasive during adolescence. Also, breakfast skipping was more closely related to BD and dieting than to family income, suggesting school lunch programs targeted to adolescents from low-income families should inform all adolescents about the risks associated with unnecessary dieting.

T19. "Matt Stone looks just ravishing in his pink dress" Irony in computer-mediated conversations.

Jeff Hancock and Philip Dunham

The text-based conversations that occur in email and internet chat environments eliminate many of the non-verbal signals, such as facial expressions and tone of voice, that we rely on to interpret a speaker's intent. The present study assessed the impact of computer-mediated communication (CMC) on the use of ironic forms (e.g., irony, hyperbole, understatement, rhetorical questions) in conversations between strangers. Given the reduction in the cues available in CMC to signal irony, we expected less use of irony in this impoverished environment than in face-to-face conversations. Comparison of 40 conversations across these two communicative environments indicated that, contrary to initial expectations, substantially more ironic language was produced in the CMC condition. This general finding has several implications for current theoretical accounts of the comprehension and production of verbal irony.

T20. Neurobiology of sleep deprivation effects on mood in major depressive disorder

Denise Bernier

The goal of this pilot study is to assess the neurotransmitter changes in the brain that correlate with the improvement of mood in depression. Sleep deprivation therapy has consistently shown an acute, although transient, improvement of depressive symptoms in more than half of patients suffering from major depression. A test-retest magnetic resonance spectroscopy will be performed before and after half a night of sleep deprivation in a sample of female patients (n=12) in their twenties, suffering from major unipolar depression. We will investigate the medial prefrontal cortex, which was already identified by brain imaging studies as showing metabolic differences before and after sleep deprivation in responders, and the limbic structures, which have been shown to be involved in emotional expression. We expect the glutamatergic peak on the MRS spectra to change in association with changes in mood.

T21. Using Taste Reactivity to Measure Shifts in the Palatability of Conditioned Taste Cues

Catherine A. Forestell and Vincent M. LoLordo

Tastes become more preferred after they have been paired with sugar reinforcement in a classical conditioning paradigm. The mechanism involved in the formation of conditioned taste preferences is not clear. It is possible that conditioned preferences occur because animals learn to expect calories upon consumption of the taste cue. Alternatively, the conditioning procedure may produce an increase in the palatability of the taste cue. Typically the taste reactivity test is used to assess the palatability of tastes. This test involves measuring a rat's hedonic and aversive affective responses to oral infusions of taste solutions. In this talk I will describe some of the taste reactivity experiments currently being conducted in our laboratory to assess palatability shifts in conditioned taste cues.

T22. Challenges of running research in a large laboratory class: An animal experiment where the researchers outnumber the animals by 30 to 1

Heather Schellinck(1), Ron Hoffman(1) and John MacDonald(2). (1) Department of Psychology
(2) Academic Computing Services

As part of the continuing evolution of the introductory psychology class (P1001) into a laboratory class, this past year an animal research project was incorporated into the curriculum. Eighteen mice were assigned to either of two groups. Mice in the experimental group received caffeine in their drinking water. The mice were then observed in an open field apparatus, and experimenters recorded several behaviours. With over 600 potential experimenters, the logistics were challenging. In this talk, we describe how live observations were combined with additional observations recorded on digital video and presented over the internet. The preparations were not trivial, but the outcome was a success in several ways.

T23. Construction and Initial Validation of a Daily Version of the Fatigue Impact Scale (D-FIS)

John Fisk and Susan Doble

Introduction: The Fatigue Impact Scale (FIS) has been developed as a symptom-specific profile measure of health-related quality of life (HRQoL). Its item content reflects subjects' attribution of problems to fatigue in the areas of physical, cognitive, and psychosocial functioning. The FIS was initially developed for cross-sectional studies of medical conditions in which fatigue is a chronic symptom. Thus, it had not been designed to be a responsive measure of daily changes in fatigue. The present study describes the development and initial validation study of an adaptation of the FIS for use in conditions in which fatigue symptoms are of acute onset or have significant daily variation.

Methods: Items for the Daily Fatigue Impact Scale (D-FIS) were selected from the pool of original FIS items with the aid of Rasch analyses of existing data. The reduced, 18-item FIS was administered to a sample of 93 subjects with flu-like illness, some of whom were followed for a 21-day period. Rasch analyses were used to further reduce the scale to a minimum number of items that represented a unidimensional measure of self-reported fatigue impact.

Results: The resulting 8-item D-FIS demonstrated good relations to ratings of other flu symptoms and to general ratings of HRQoL. It also proved to be a responsive measure, demonstrating significant reduction in fatigue impact for those subjects that were followed longitudinally. Moreover, subjective ratings of fatigue impact were associated with work hours lost and estimated work productivity for those subjects who did work for pay during the study period.

Conclusions: This initial validation study indicates that the D-FIS has considerable promise as a measure of the subjective daily experience of fatigue and as a specific HRQoL outcome measure.

Handwritten notes:
 Several studies recently of ...
 The instrument was ...
 ...

T24. The Concordance Between Health-Related Quality of Life Utility Measures and Neurologic Disability Ratings in Multiple Sclerosis

John Fisk, Murray Brown, Luanne Metz and T.J. Murray

Introduction: Clinical trials of disease-modifying agents for MS have focused on neurologic disability, relapse rates, and MRI measures of central nervous system lesions. While the results have been promising, study limitations, coupled with the high costs of treatments and the limited evaluations of their cost-effectiveness, have lead to controversies about their use. The goal of this study was to evaluate the relation between clinical ratings of neurologic disability and generic Health- Related Quality of Life (HRQoL) utility measures in MS patients. *Methods:* A total of 195 patients with MS (47 males, 148 females), representing the full range of neurologic disability (i.e. no neurologic signs to bedridden) were recruited in a stratified (neurologic disability) and randomized manner from two regional MS clinics. All subjects received a standardized clinical assessment and completed the EuroQol (EQ5D) and the Health Utility Index (HUI) Mark II and Mark III. *Results:* The EQ5D and both versions of the HUI all demonstrated correlations $> .60$ with the clinical ratings of neurologic disability. Analyses of variance demonstrated that the HUI Mark II and, to a greater extent the HUI Mark III, were able to distinguish between clinically distinct subgroups of patients. A surprising finding was the marked differences in the extent and rate of decline of the various HRQoL utility measures across the full range of increasing neurologic disability. *Conclusions:* HRQoL utility measures can be appropriate outcome measures in clinical trials for MS. Differences in item content and in the valuation methods are evident in how the subjective health states of a specific disease population are represented by HRQoL utilities. The selection of a HRQoL utility measure can have a profound effect on the cost-utility ratios in pharmacoeconomic analyses of MS treatments.

T25. Patterns of Orexin Immunoreactivity in the Brains of Alzheimer's Disease Patients

Rusak, B., Darvesh, S., Hopkins, D., Goguen, D., Fice, D., Kale, K., Reid, A. and de Groot, M.
Departments of Psychiatry, Medicine (Neurology), Anatomy/Neurobiology and Psychology

Orexins (hypocretins) are a class of peptides originating in cells of the hypothalamus that are attracting great interest. Orexins appear to play a critical role in regulating arousal; for example, loss of either orexins or their receptors have been implicated as a cause of narcolepsy in animals and humans. In addition, orexins have an important impact on food intake in animal models. Patients with Alzheimer's Disease (AD) often show clinical symptoms related to dysregulation of both sleep and body weight. While neuronal cell loss occurs in many brain regions in AD patients, the regions responsible for dysregulation of feeding and sleep are not known. We have conducted immunocytochemical studies of selected regions of the brains of AD patients and age-matched controls to evaluate the number of orexin-immunoreactive neurons and the extent and pattern of orexin fibre staining. Preliminary results will be presented.

T26. Mousing around with sensory preconditioning

Sarah E. Boudreau and Vincent M. LoLordo

The ability of mice to acquire a stimulus-stimulus association was tested with odor stimuli in a sensory preconditioning paradigm, as a means of measuring their abilities to integrate separately learned associations. Two distinct odor stimuli, rose and lemon, were presented in buried odor pots. On the first day of training mice in the paired group received paired presentations of odor 1 (O1; lemon) and odor 2 (O2; rose). Mice in the unpaired group received randomized odor presentations 15 minutes apart. On the following day, both groups received paired presentations of O2 and a sugar reward. On the third day, both groups of mice received one trial of O2 and sugar followed immediately by trials of O1 alone. If separate associations were learned and integrated, then those in the paired group should spend more time digging in O1 on the final day than mice in the unpaired group. Results to be discussed.

T27. Plasmid protection rackets

Will Brown and Ford Doolittle

Bacterial plasmids are self-replicating elements that exist within a cell but are not part of the chromosome proper. When plasmids are mobile they can laterally transmit between phylogenetically distant lineages (e.g. *E. coli* and *S. enterica*) and confer advantages to the host organism. For example antibiotic resistance is probably a case of adaptive host evolution due to lateral transfer. However, why do plasmids transmit themselves laterally? Are host benefits a sufficient explanation to explain the presence of plasmids? Benefits to the host could be an epiphenomenon of benefits to the plasmid. It is suggested that social evolution may favour multi-level 'conspiracies' between evolutionary units and this potentially leads to conflicts within a given level of selection (e.g. plasmid 'protection rackets'). Despite the effectiveness of conspiratorial selection between evolutionary units, the threat of mutinous counter-adaptations may exist in all social systems (e.g. meiotic drive, maternal-effect selfish genes or postsegregational killing in plasmids). It seems that, in contrary to intuitive 'folk wisdom,' the brain is not necessary for social evolutionary processes to operate.

T28. The influences of increased working memory load on semantic neural systems.

D'Arcy, R.C.N.(1), Connolly, J.F.(1,2), Service, E.(3), Hawco, C.(1), & Houlihan, M.(4)

1) Cognitive/Clinical Neuroscience Unit, Department of Psychology, Dalhousie University, Halifax, Canada; 2) Department of Psychiatry, Dalhousie University, Halifax, Canada; 3) Department of Psychology, University of Helsinki, Helsinki, Finland; 4) Pediatric Pain Laboratory, IWK Grace Health Centre, Dalhousie University, Halifax, Canada

The effects of WM on the semantic N400 response were studied using high-resolution ERPs. Subjects studied visual sentence primes (The man is teaching in the classroom), anticipated the superordinate terminal word (school), and then evaluated the semantic appropriateness of the visual target sentence (The man is in the...school/barn). Level 1 WM load contained a single prime sentence and Level 2 contained two prime sentences. The inappropriate words elicited the N400, but the latency of this response was delayed significantly by ~50ms due to increased WM load. Subsequent brain mapping and source analysis techniques showed the effects of WM on N400 generators.

T29. More on Marleen's Misbehaving Mice: Maximizing the Mating of Monogamous Mutants

Marleen H. M. de Groot & Benjamin Rusak

California mice (*Peromyscus californicus*) are studied by Richard Brown's lab because of their unusual pattern of bi-parental care of the young. We became interested in these mice because, unlike other species we have studied, they show high levels of activity in their home cages in anticipation of the onset of the daily light phase. Although the circadian systems of two species of *Peromyscus* mice have previously been studied (*P. leucopus* and *P. maniculatus*; Daan & Pittendrigh, 1976), circadian rhythms of California mice have not been described in the literature. Eighteen adult male California mice were housed in individual cages inside isolation boxes that allow for manipulation of each animal's light exposure. Activity rhythms were recorded under a variety of lighting conditions. Although most of these mice behaved similarly to other *Peromyscus* species studied previously, a few animals showed extremely unusual patterns. Efforts to further characterize these anomalous patterns, and to selectively breed for unusual circadian characteristics will be discussed.

T30. Relation between catastrophizing, depression, and the intensity of cold and influenza symptoms

Devoulyte, K., Laskis, S., & Sullivan, M.

The current study examined the role of catastrophizing and depression in predicting the intensity of cold and influenza symptoms. Participants included 49 adults (35 females, 14 males) who agreed to complete a diary of symptoms for 7 days after the onset of a cold or the flu. Participants were classified as catastrophizers or noncatastrophizers based on their scores on the Pain Catastrophizing Scale (PSC). Results showed that catastrophizers reported significantly more symptoms, and significantly more severe symptoms than noncatastrophizers. The relation between catastrophizing and symptom severity remained significant even when controlling for initial symptom severity, duration of onset of symptoms, and depression. Catastrophizers also reported using more medication than noncatastrophizers. Catastrophizers did not differ from noncatastrophizers on an objective measure of fever, nor did they differ in the amount of school/work time lost due to symptoms. The theoretical and practical implications of these results are discussed.

T31. The Development of the Objective Self

Nancy Garon and Chris Moore

The present study explored the development of the objective self. This refers to the child's ability to perceive the self from a third person perspective. Other measures of this ability such as the mirror recognition task have been related to important developmental changes. It was hypothesised that this ability develops in stages consistent with the development of other cognitive abilities. Two tasks, thought to measure this ability, were created for this study. The first task assessed children's ability to reason about their mass while the second task assessed children's ability to reason about their size. Children in different age groups showed significant differences in performance on both tasks. The results of this study suggest that a child's ability to represent the objective self changes considerably over the preschool period.

T32. Procedures for testing the sensory abilities of inbred, transgenic and mutant mice: a review of the literature

Guy Gingras and Richard E. Brown

With the increasing number of strains of inbred, transgenic and mutant mice available, it is common to use these mice in studies of cognitive functions. However, many strains of mice have sensory deficits and it is important to test for sensory function before using these mice in tests of cognitive function. A number of review papers have been written on the behavioral phenotypes of mutant mice, but unfortunately, little information is available on the sensory capacities of these mice. Moreover, online databases on mutant mice are rather broad and contain little information on behavioral abnormalities. Because behavioral tests are dependent on sensory abilities, basic information on sensory abnormalities in mutant mice is important for research in the neural and the behavioral biology of the mouse. The goal of this presentation is to describe the methods used in testing the sensory functions (vision, hearing, taste, smell, touch, and pain) of normal and mutant mice. To our knowledge, this is the first attempt to create a comprehensive review of the sensory abilities of mice.

T33. ACh is AChing for Attention

Doug Rasmusson and Kazue Semba, Depts. of Physiology & Biophysics and Anatomy & Neurobiology

Acetylcholine (ACh) is important because its release in the cortex is high during EEG "activation", it enhances sensory input and its neurons are lost in Alzheimer's disease. However, we don't understand if this cholinergic system functions by activating the entire cortex in preparation for significant inputs or if separate cortical regions can be activated independently.

This is partially an anatomical question (do individual neurons branch to many regions or do different cholinergic neurons go only to each region?) that will be examined using tract-tracing techniques.

It will also be examined functionally by measuring ACh release from two cortical areas under different sensory conditions-can ACh be released selectively from one region and not from another? or does it increase everywhere regardless of the input? This will be studied in anesthetized animals using visual and auditory stimuli and then in awake behaving animals trained to carry out difficult visual or auditory discriminations.

T34. Why psychology isn't a science.

John Chrstie

T35. TBA

J. Blustein

T36. How does the circadian clock tell the rest of the brain it's time to wake up?

Kazuo Semba, Samuel Deurveilher and Joan Burns, Departments of Anatomy & Neurobiology, and Psychology.

Daily cycles of sleep and waking characterize the lives of various species. In mammals, the master clock for these circadian rhythms is located in the suprachiasmatic nucleus (SCN), whereas the neuronal system controlling sleep and wakefulness is distributed throughout the brain. To understand how these two systems are linked, we previously electrophysiologically characterized direct projections from the SCN to the ventrolateral preoptic area (VLPO), a major site of sleep-active neurons (Sun, Whitefield, Rusak and Semba, submitted). However, the most of the SCN output is thought to reach its targets indirectly. In the present study, therefore, we are using a cocktail of anatomical tracers to identify multisynaptic SCN pathways. The potential relay nuclei so far identified include the medial preoptic area, dorsal hypothalamic area, and dorso- and ventromedial hypothalamic nuclei. These nuclei could relay SCN output to the VLPO, and to known wake-active cell populations, including neurons containing hypocretins/orexins, histamine, dopamine, acetylcholine, serotonin and noradrenaline in the forebrain and brainstem.

T37. Nonpredictive Gaze and 14-Month-Old Infants: Saccadic Responses to Static and Animated Cues Varying in Validity and SOA

Amy Carr

Reflexive orienting to nonpredictive gaze cues has been demonstrated in adults, young children, and infants (Friesen & Kingstone, 1998; Hood, Willen, & Driver, 1998; Ristic, Friesen, & Kingstone, in preparation). In the present study, as in the Ristic et al study, a schematic happy face was used to present nonpredictive central gaze cues. Infants' saccadic responses to subsequent peripheral targets were examined. Out of 37 infants tested, 25 (mean age: 14 months; 9 days) met the inclusion criteria of 8 usable trials. The design was mixed, with cue type (static or animated) as a between-subjects variable, and cue validity (valid or invalid), target location (left or right), and stimulus onset asynchrony (200 ms, or 1000 ms) as within-subjects variables. At the present stage of analysis, none of the anticipated effects are significant. A follow-up study will use more interesting and age-appropriate stimuli to examine the effects of nonpredictive gaze cues.

T38. Effects of the Swiss Air Flight 111 disaster on community-level alcohol consumption: Post-traumatic drinking?

Sherry H. Stewart and the "Learning from Experience" Research Team

BACKGROUND: The previous literature is controversial whether communities respond to disaster with increased or decreased alcohol consumption. Inconsistencies may stem from the different time periods following disaster over which communities are tracked. For example, decreased drinking may occur during the initial response to disaster, when communities are mobilizing their resources to deal with the trauma. Increased "post-traumatic drinking" may still occur during later stages of a community's response to disaster, as some members attempt to cope with symptoms of post-traumatic stress. By examining liquor sales over a wider time span, this study may help bring resolution to prior inconsistencies. **OBJECTIVE:** To determine whether there are changes in alcohol sales in coastal communities affected by the 1998 airline disaster in Peggy's Cove, Nova Scotia. **DESIGN:** Liquor sales records are being examined in communities affected by the disaster and control communities for the months and years prior to, during, and after the disaster. **SETTING AND SUBJECTS:** Liquor stores serving coastal communities affected by the disaster are being compared with liquor stores serving matched control communities. **RESULTS:** Analyses will be performed both within affected communities over time, and between affected and control communities. We examine whether alcohol sales increased or decreased in the period immediately following the disaster, whether there is sustained increased sales over time, or a delayed increase in alcohol sales well after the disaster. **IMPLICATIONS:** The results could help identify whether alcohol abuse prevention needs to be provided for communities affected by large-scale disaster.

T39. What does the human genome project mean for Psychology?

Richard E. Brown

The human genome was published in *Science* and *Nature* in February 2001. The mouse genome, *Drosophila* genome, *C. Elegans* genome and others have also been published. It is predicted that everything we know in pharmacology, medicine and human biology will have to be revised in light of genetic knowledge. How will these breakthroughs in genetics impact on psychology? Will everything in psychology have to change to reflect genetic knowledge? Do psychology students understand the language and concepts of genetics? What new theories and techniques will be required for animal research? Will genetics manipulations, gene chip technology and gene insertion by adenoviral vectors become standard tools in the study of brain and behaviour? Can we keep up with the demand for neurobiological and behavioural studies on transgenic and mutant mice? What will we have to learn about genetics to facilitate human experimental and clinical research? Finally, how can psychologists use bioinformatics and neuroinformatics, the handling and use of large genetics databases, in their research? This talk examines some of the emerging knowledge in genetics that psychologists will need to integrate into their teaching and research in order to benefit from the genetics revolution.

T40. Human Patient Simulator: A proposed feasibility study

James Clarke and Peggy Ruyak

The Human Patient Simulator (HPS) is a highly sophisticated computer based mannequin used to train a wide range of medical professionals. The HPS is extremely diverse and can provide training in acute crisis resource management, basic and advanced medical interventions and management, basic pharmacology, and equipment usage. This unique training tool has been used extensively in the training of medical personnel but it has been virtually ignored for research purposes. One of the difficulties that arise in the use of the HPS for research is the lack of standardized scenarios. The proposed study evaluating the performance of sleep deprived paramedics would allow us to evaluate the capabilities of the simulator, develop standardized scenarios, and establish a research program using the HPS as a research tool.

T41. What matters in personal survival: Philosophical questions and empirical answers.

Jennifer Theakston and John Barresi

What do people care about in personal survival? If they have to change and become different from what they are now, what do they think is most important to maintain about themselves through change? Some philosophers have suggested that what is important to maintain in survival is one's identity. But what does identity mean for a person? Others say that what matters in survival is that one can identify with the person that continues even if the person that one is now no longer exists. But who decides when a 'continuer' is no longer one's self? Rather than answer these philosophical questions the present investigation looks into which aspects of self matter to people in survival: what they care most about preserving in change, even in circumstances in which death is one the alternatives.

T42. Homicide Perpetration as a Function of Psychopathy: The Instrumental-Reactive Continuum Explored.

Mike Woodworth & Stephen Porter

The purpose of the present study was to examine the characteristics of a large sample of Canadian homicides as a function of psychopathy. Although previous studies have investigated the relationship between psychopathy and criminal behaviour, little is known about the relationship between psychopathy and homicide. One hundred and twenty-five ($N = 125$) Canadian offenders who had perpetrated a criminal homicide were investigated for several characteristics regarding the nature of, and possible motivations behind, their offence. Results indicated that homicides committed by psychopathic offenders (as measured by the Psychopathy Checklist-Revised; Hare, 1991) were significantly more instrumental (planned, goal-oriented) relative to non-psychopathic offenders who committed more spontaneous "crimes of passion." In addition, psychopathic offenders were found to commit more gratuitous, sadistic, and sexual violence in the content of their homicides than non-psychopathic offenders. This research strongly contributes to the scientific understanding of homicidal violence as a function of psychopathy, and could have important implications for homicide investigations.

T43. Investigation of the Consistency of Traumatic Memories over Time as a Function of Interview Style

Kristine A. Peace & Stephen Porter

In the legal system, it is often important to evaluate the credibility of memories for distant emotional experiences. The purpose of the proposed study will be to examine prospectively the characteristics and consistency of both traumatic and positive emotional memories as a function of interview style.

Although previous studies have investigated the qualitative aspects of emotional memories, little is known about the consistency of traumatic memories over extended periods of time. The goals of the current study are to contribute to the understanding of characteristics of traumatic memory, as well as examine the extent to which these memories are consistent over time. It is expected that this research will reveal important findings to further our understanding of how traumatic experiences are remembered, in addition to potential enhancing or biasing influences of various interview techniques.

T44. Who Can Catch a Liar? An Investigation of the Effect of Motivation and Feedback on Deception Detection

Sean McCabe, Stephen Porter, Angela R. Birt, & Mike Woodworth

Previous research indicates that most people are not overly proficient at detecting deception, typically performing around the level of chance. However, a recent study suggested that the provision of immediate and accurate feedback may gradually facilitate the accuracy at detecting deceptive reports (Porter, Woodworth, & Birt, 2000). This apparent improvement in accuracy from the feedback could be related to either: (1) an increase in the social pressure from the feedback to elevate the participants' motivation to make accurate judgments, or (2) the feedback may have allowed the participants to gradually adopt more successful decision-making strategies. The present experiment was designed to examine the validity of these competing theories. Participants ($N = 151$) were randomly assigned to two experimental conditions according to motivation (High versus Low) and feedback (Accurate, Inaccurate, or No Feedback). They were shown a series of twelve videotaped speakers providing either a true or deceptive report. Contrary to the major hypothesis, results indicated that the highly motivated participants (those who anticipated a reward for performing well) performed significantly less accurately than low motivated participants (who were not motivated by or aware of any possible reward). A significant effect for any feedback (no difference for accurate versus inaccurate) was found for the accuracy for the honest videos and a significant interaction between motivation and feedback was seen in the accuracy for deceptive targets. Although results indicated a general motivational impairment effect for the detection of deception, any type of feedback appears to counteract this impairment effect and facilitate judgements on liars.

T45. Are Left Handers Better Lie Catchers? An Investigation of Individual Differences and Accuracy in Deception Detection

Paul Angelopoulos, Angela R. Birt, Stephen Porter, & Mary Ann Campbell

In a recent study (Porter, Stapleton, Campbell, & Birt, 2001), it was found that left-handed individuals were able to detect deception at a higher level than right-handed individuals. The present study was intended to replicate this result and explore possible reasons for the difference. Participants ($N = 110$) viewed twelve videotaped targets (six liars, six truth-tellers). They were asked to judge the honesty of the video targets. The videos were presented in one of three modalities: audio only, visual only and audiovisual. Participants were asked to complete a number of questionnaires to explore possible information processing styles associated with accuracy in detecting deception. It was hypothesized that left-handed participants rely more on nonverbal information such as facial cues when making credibility judgements, and therefore, would have higher accuracy levels in the visual only and audiovisual conditions than right-handed participants. Overall, left-handers were 14.17% more accurate in judging deception than were right handers. As hypothesized, left handers displayed this superiority effect in both the visual and audiovisual conditions. No significant differences in accuracy scores between left and right handers were found in the audio only condition. Analysis of self-reported cues upon which detection judgements were based revealed a trend for left handers to rely primarily upon cues of a visual nature. The findings are discussed in relation to possible differences in information processing tendencies/styles between left and right handed individuals.

T46. "Bleeding Men and Parrot Ladies": Effect of emotionality of a scene on susceptibility to false memories

Lesley Spencer, Stephen Porter, & Angela R. Birt

Although some level of distortion is commonly present in eyewitness memory, the relation between the emotionality of a witnessed scene and susceptibility to false memories is less clear. In the present investigation, participants ($N = 90$) were randomly assigned to one of six conditions in a 2 (misled versus non-misled) X 3 (positive, neutral, or negative emotional scene) between-subjects design. Participants were informed that the study was focusing on "emotional processing" and were not aware that their memory for the scene would be tested. After viewing a highly positive, neutral, or highly negative scene (e.g., graphic accident) from the International Affective Picture System (Lang & Ohman, 1988), half of all participants were exposed to a series of misleading questions, one of which included a dramatic false suggestion (e.g., a large animal in the scene that did not exist). Approximately one hour later, after a series of filler tasks, participants were given ten direct questions about the scene (five related to the misinformation provided earlier). Results indicated that the misleading questions had a substantial impairing effect (37% difference) on recall accuracy. Overall, emotional valence did not significantly influence susceptibility to memory distortion. However, it was related to false memories for the dramatic misinformation in misled participants. Specifically, misled participants in the negative emotion condition recalled seeing the dramatic misinformation significantly more often (80%) than participants in either the positive (40%) or neutral emotion (40%) conditions. Further reflecting the strong impact of misleading questions, no (0%) non-misled participants in any condition recalled seeing the dramatic misinformation. Possible implications for eyewitness memory are discussed.

T47. Young Offender Recidivism: Development and Validation of a Comprehensive Risk Scheme (A Doctoral Dissertation Proposal)

Mary Ann Campbell & Stephen Porter

Psychopathy is over-represented in criminal offender populations, is associated with persistent and violent offending, and is considered an important component of adult risk assessment (Hare, 1998; Hemphill, Templeman, Wong & Hare, 1998; Salekin, Rogers, & Sewell, 1996; Serin, 1995). Although psychopathic traits have been identified in both children and adolescents (Forth, Hart & Hare, 1990; Frick, O'Brien, Wootton, & McBurnett, 1994), the relevance of psychopathy to young offender recidivism risk is less clear. The main objectives of the proposed research will be to: 1) determine the base rate of psychopathy in a large sample of young offenders; 2) retrospectively identify the psychosocial and criminal correlates of youth psychopathy in order to identify potential early markers in its development; 3) evaluate the contribution of an adolescent measure of psychopathy to the prediction of past and future criminal behaviour; 4) identify the criminal and psychosocial factors discriminating persistent offenders high in psychopathic traits from non-persistent psychopathic youths; and 5) develop a much needed comprehensive risk/need assessment instrument based on psychopathy and other factors contributing to the prediction of re-offending in young offenders. To meet these objectives, the psychosocial and criminal history data previously collected from 227 male and female young offenders (see Santor and Kusumakar, 1998) will be used. The PCL-YV (Forth, Kosson, & Hare, in press) will be scored from this information to measure psychopathy. Recidivism will be measured from the point of participation in the original study to the present (3-4-year follow-up). Half of the sample will be used to develop a risk/need measure, while the other half will be used in a cross-validation. Collectively, these results will contribute to the understanding of psychopathy in its earlier stages and provide a useful tool for the assessment of risk/need in young offenders.

P1. Changes to the object recognition system in patients with dementia of the Alzheimer's type

Kerri Purdy, Paddy McMullen and M. Freedman

Do patients with Dementia of the Alzheimer's Type (DAT) show category-specific deficits in object identification, and if so do they arise from semantic or visual damage? Basic (e.g., dog) and subordinate (e.g., collie) level identifications require a large amount of visual processing, whereas superordinate (e.g., animal) level identification requires additional semantic processing beyond the basic level. DAT patients, normal elderly controls and college-aged controls completed an object verification task, where they decided whether line-drawings of living and non-living objects matched names at superordinate, basic or subordinate levels. Superordinate and subordinate identifications were equally difficult for DAT patients. Subordinate identifications were most difficult for all controls. A category-specific deficit for living objects was found with DAT patients, but only at the Superordinate level. This finding supports the notion that some differences between living and non-living objects have a semantic basis. These findings also characterize the nature of the semantic memory deficit in DAT.

P2. A comparison of HS mice in the Barnes Maze and the Morris Water Maze

Darryl Houghton, Lianne Stanford & Richard E. Brown

The Morris water maze, which is frequently used as a test of learning and memory in rodents has been criticized as being a stressful paradigm in mice (Van der Staay, 2000: *Physiol. Behav.* 68, 599-602 & Brooks et al. 2000: *Soc. Neurosci Abst.* 30, #841.3). This stress may confound learning and memory. The Barnes maze (BM), a visual spatial task in which mice learn to escape from the center of a brightly-lit platform into a dark cage through 1 of 16 holes on the periphery, may be a less stressful alternative to the MWM. If these two tasks are accessing similar cognitive processes, measures of learning and memory in the MWM should be highly correlated with those of the BM. To test this hypothesis, we ran 34 Heterozygous Stock (HS) mice (IBG, CO, USA) in both the MWM and the BM. Mice in the MWM showed decreased escape latencies (EL) across days and increased EL on the first day of reversal. Mice in the BM had decreased EL in acquisition, but no increase in EL on the first day of reversal. The number of errors committed in the BM decreased over days and did increase on the first day of reversal. Escape latency and errors within the BM correlated for all measures of learning ($.45 < r < .92$, $p < .0001$ to $.0072$). Separately, each of these tasks has measured learning and memory. In comparing the two tasks, correlations for the escape latencies of the BM and MWM were calculated and all measures of learning [all days combined ($r = -.24$, $p < .1969$), day - one acquisition ($r = -.03$, $p < .8775$), all acquisition ($r = -.13$, $p < .4822$), day-one reversal ($r = .13$, $p < .4686$), & last day of reversal ($r = -.13$, $p < .4568$)] or memory ($r = .05$) were not significant. There were also no significant correlations between the BM (errors) and the MWM (EL) for any measure of learning [all days combined ($r = -.15$, $p < .4011$), day-one acquisition ($r = -.01$, $p < .9545$), all acquisition ($r = -.13$, $p < .4965$), day-one reversal ($r = -.18$, $p < .3004$), & last day of reversal ($r = .04$, $p < .8229$)] or memory ($r = .10$ for errors). These data suggest the cognitive processing for these two tasks may be different, and when comparing data between these tasks caution should be exercised.

P3. Information provided by skeletal muscle and lower sensory/motor neurons influence neuronal fate in the developing mouse brain

Boris Kablar and Michael A. Rudnicki

Fetuses that develop without Myf5 and MyoD genes entirely lack skeletal myogenesis and lose spinal motor and proprioceptive neurons. To test whether extrinsic cues provided by the lacking structures influence neuronal precursor cell differentiation in various brain areas involved in motor control we assessed wild-type and mutant brains for structural and molecular differences. Anatomical milestones of the mutant brain appeared unaffected. However, the expression pattern of various molecular markers for neural progenitors and pyramidal neurons in the mutant motor cortex was downregulated. Consistently, some areas of the mutant extrapyramidal tract were also affected. Surprisingly, we were unable to detect any change in proliferation or cell death in the mutant neuroepithelium. Taken together, it appears that patterning mechanisms intrinsic to the areas for motor control specify the basic organization of its subdivisions. However, extrinsic cues seem to change intrinsic properties of neural progenitors and influence the ratios of the cell types they produce.

P4. Linking neuropsychological and neuropsychological assessments

Yannick Marchand, Ryan D'Arcy & John Connolly

There is considerable potential for the use of event-related brain potentials (ERPs) for neurocognitive evaluation in a clinical setting. The objective of the current study was to evaluate a group of left-sided stroke patients with varying degrees of receptive language impairment using both neuropsychological testing and ERP assessment. It was hypothesized that ERPs will assess accurately receptive language function and correlate with the neuropsychological data. Data were collected from ten left-sided stroke patients, who were undergoing rehabilitation at the time of testing. ERPs were recorded during the computerized Peabody Picture Vocabulary Test-Revised (PPVT-R), which has graded levels of vocabulary difficulty. In the computerized PPVT, a picture is presented followed by a digitized spoken word that is either congruent or incongruent with the picture. The results provide a comprehensive demonstration of the efficacy of ERP neurocognitive assessment in a stroke population.

P5. Induction of Hsp27 in the cerebellum after focal cerebellar injury

Allen, G.V., Chase, T., Sisca, L. and Reynolds, L. Anatomy & Neurobiology

Heat shock protein induction is thought to play a role in neuroprotection after traumatic brain injury (TBI) or ischemic insults. Induction of the 70 kDa heat shock protein has been described in Purkinje cells following various types of brain injury. However, information regarding expression of the small 27 kDa heat shock protein (Hsp27) after focal cerebellar injury is lacking. In this study, we used a weight drop model of brain injury in the anesthetized rat to identify tissue damage and Hsp27 induction. Male, Sprague Dawley rats were anesthetized with isoflurane and a craniotomy (5.0 mm) was made over the left paravermal cerebellar cortex using manual manipulation with an auger drill bit. The bone flap was removed (dura intact) and brain injury was produced by dropping a 20-g brass weight (severe TBI=15 cm height, 300g/cm, mild TVI=2.5 cm height, 50g/cm) down a plexiglas guide tube onto a plexiglas piston (3.5 mm diameter). Immediately after trauma, the bone flap was replaced and sealed with dental cement. Control animals were treated the same except that they did not received a weight drop injury. The animals were killed at different time intervals after injury and brain sections were processed for the immunohistochemical detection of Hsp27 or stained with thionin. Hsp27 is not constitutively expressed in the cerebellum. Following mild or severe weight drop injury, Hsp 27 induction was observed in Purkinje cells and astrocyte-like cells located in the molecular and granular cellular layers. Hsp27-positive astrocyte-like cells were prominent in widespread regions of the cerebellar fiber tracts predominately ipsilaterally. Following severe TBI only, Hsp27 labeled neurons and glial cells were observed in the deep cerebellar nuclei. Thionin staining showed prominent chromatolysis in the Purkinje cell layer directly under the TBI site and gliotic scarring in the deep fiber tracts ipsilateral to the injury. The data demonstrates that induction of Hsp27 occurs in both cerebellar neuronal and neuroglial cell population s after traumatic cerebellar injury. The role of HSsp27 induction in promoting cell survival or providing protection against subsequent injury needs to be determined.

P6. Local Attentional Deficits, Prosopagnosia and Category-Specific Agnosia for Living Objects

Paddy McMullen, Amelia Hunt and Kerri Purdy

M.B., a 41 year old prosopagnosic with category-specific agnosia for living objects, was tested 10 years after her original testing session for her ability to 1) detect configural and element-based changes in schematic faces and non-faces, 2) process at global and local attentional levels and 3) verify the identity of living and non-living objects at different levels of categorization. In the face task, normal controls were more sensitive to element changes in faces and non-faces than to configural changes. M.B. showed equivalent and low sensitivity to both element and configural changes in faces. MB was much more sensitive to changes in nonfaces, with a normal pattern of greater sensitivity to elemental changes. She was also impaired at local processing of Navon-type figures in a divided attention task. Finally, normal controls showed a living advantage and more efficient decisions to basic names than subordinate or superordinate names when superordinate, basic or subordinate names of living and nonliving objects were matched to pictures. M.B. showed very deficient processing of subordinate names in this task. Interestingly, her category-specific deficit was not manifest 10 years after her original deficit was found. M.B. has a visual impairment that manifests as a general face-processing problem that may be linked to abnormal local attentional allocation. Her category-specific problems were likely of a more semantic nature.

P7. Long-Term Kindling in Mice as a Model for Interictal Emotionality

A.C. Davis, C. Dodd, N. Young, & E. G. Marchant

Patients with temporal lobe epilepsy (TLE) not only suffer from recurrent spontaneous seizures but also show disturbances in interictal (between seizures) emotionality, i.e. depression and anxiety. Long-term kindling is an animal model that is used to study these increases in emotionality. Most of the research to date on kindling-induced emotionality has been done with rats, and none has been done with mice. The advantage of using mice in this model is that the use of transgenics may allow for a greater understanding of the neural mechanisms underlying long-term kindling-induced emotionality. The purpose of this study was two-fold: to determine the effects of kindling on emotionality in mice, and to determine a technique that would allow for a maximum survival rate in mice for the duration of the long-term stimulations. Mice were given either 30, 75 or sham stimulations, and were tested on the open field and elevated plus maze. Significant differences between kindled and control mice were found in time in the periphery of the open field, line crosses in five minutes in the open field and open arm entries on the elevated plus maze. Also the survival rate of the mice for the duration of the experiment was 78%. The results from this experiment suggest that kindling in mice does produce changes in emotionality and that it is possible to deliver long-term stimulations to mice. These results suggest that future studies using transgenics in the long-term kindling model may be possible.

P8. Criminal Behaviour and Release Performance in Psychopaths and Non-psychopaths across Adulthood

Angela R. Birt & Stephen Porter

Using the Psychopathy Checklist-Revised (PCL-R; Hare, 1991) diagnostic cut-off score of 30, the complete criminal career and community release profiles of 317 Canadian federal offenders (224 low scorers and 93 within the psychopathic range) were investigated retrospectively. Adult crimes were coded according to age at commission as well as either violent, non-violent, or non-sexually violent. Changes in performance following release into the community also were examined. Results indicated that offenders scoring within the psychopathic range consistently committed more violent and non-violent crimes than their counterparts for about three decades, spanning their late adolescence to their late 40s. Numbers of non-violent criminal offenses committed by high PCL-R scorers declined considerably after age 30 relative to violent offenses, which declined and then rebounded in the late thirties before a major reduction was evidenced. Throughout adulthood, high PCL-R scorers failed during community release significantly faster than did low PCL-R scorers. Importantly, from a risk management perspective, the release performance of low PCL-R scorers improved with age, while it worsened for high scorers as they got older. Further, those offenders scoring high on the PCL-R did not show a lower charge to conviction ratio with age, suggesting that they were not getting better at manipulating the legal system.

P9. "If I Didn't Know Better, I'd Think You Were Lying to Me...": Influence of Pre-judgment Biases on Credibility Judgments"

Naomi Doucette, Angela R. Birt, Stephen Porter

This study examined the effect of judgment bias on the accuracy in detecting deception. One hundred and two participants judged the veracity of 12 video taped statements (6 honest and 6 deceptive) and were asked to make ratings of credibility. Participants were randomly assigned to one of three conditions: no bias, honesty bias, or deceptive bias. In the deceptive bias condition, participants were instructed that it was quite possible that the speakers in the videos were attempting to be deceptive. By contrast, participants in the honesty condition were instructed that the speakers were likely to be telling the truth. Participants in the no bias condition were led to believe that the purpose of the study was to simply judge the quality of memory reports and no mention of deception or honesty was made. Overall accuracy in detecting deception was at chance level across the three conditions. However, the instructional manipulation did bias judgments and the degree of bias depended on the nature of the instructions. The participants in the no bias condition were significantly less accurate than those participants asked to focus on either honesty or deception. The frequencies with which deceptive and honest judgments were made also varied across the conditions. The implications of such pre-judgment biases in the context of legal and forensic settings are discussed.

Changes to In-House Program

- Richard Brown will include a reminiscence about Vern Honig's life in the department during his introduction to the In-House.

- Additional poster for Thursday, April 26, 2001:

Switch costs without task-switching? An inter-trial examination of the Simon effect

Jason Ivanoff & Amelia Hunt

In three experiments we examined whether switching the task-irrelevant mapping between stimulus and response results in a task-switch cost. Contrary to theories of task-switching that maintain that the switch cost is solely due to task-set re-configuration, a cost was observed.

- Ben Rusak will be the presenter for **T25) Patterns of Orexin Immunoreactivity in the Brains of Alzheimer's Disease Patients.**
- John Barresi will be the presenter for **T41) What matters in personal survival: Philosophical questions and empirical answers.**
- T22) Heather Schellinck, Ron Hoffman and John MacDonald will be switching times with T37) Amy Carr.
- And last but not least, a typo on Friday, April 27, 2001. Lunch will be from 12:30 to 2:00 pm

To: Suzanne King <Suzanne.King@dal.ca>
From: rebrown@is.dal.ca (Richard Brown)
Subject: Re: InHouse Programs

23 April 2001

The In-House programs are ready and available to be picked up in the main office. The charges this year will be as follows:

Faculty, Postdocs and Staff \$16.00
Graduate and Undergraduates \$10.00

These fees cover your program, lunch (both days), coffee breaks and pizza on Friday night.

We are asking people to try and pay cash since we no longer have a Psychology Fund (Coke Fund). There will be NO IOU's.

Thanks.

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Suzanne King Phone: (902) 494-2262
Administrative Secretary FAX: (902) 494-6585
Department of Psychology Email: Suzanne.King@dal.ca
Dalhousie University website: www.dal.ca/Psychology
Halifax, Nova Scotia
Canada B3H 4J1

To: Suzanne King <Suzanne.King@dal.ca>
From: rebrown@is.dal.ca (Richard Brown)
Subject: Re: Addendum to In-House Program

Date: Mon, 23 Apr 2001 16:21:25 -0300
From: Suzanne King <Suzanne.King@dal.ca>
Subject: Addendum to In-House Program
X-Sender: suzannep@is.dal.ca
To: neuro-list@ac.dal.ca
MIME-version: 1.0

Content-type: text/plain; charset="us-ascii" ; format="flowed"

1) Richard Brown will include a reminiscence about Vern Honig's life in the department during his introduction to the in-house.

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Thanks.

